K = simplicial complex

$(G, +)$ = group ← use this as a coef. group

Define:

an i-chain is a sum of i-simplices, using coeficients in G

$$C_i = \left\{ \sum_{\sigma \in K_i} \alpha_\sigma \sigma \;\middle|\; \alpha_\sigma \in G \right\}$$

K_i: the set of i-simplices (with orientation)

α_σ is a coef. here. G tells us how to "add" simplices. eg. If $\alpha, \beta \in G$, $\alpha\sigma + \beta\sigma = (\alpha+\beta)\sigma$

Boundary:

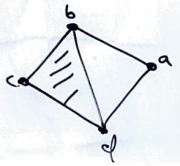
$$\partial_i : C_i \longrightarrow C_{i-1}$$

$$\sigma = [v_0, v_1, \dots, v_i] \in K_i \longmapsto \sum_{j=0}^{i} (-1)^j [v_0, v_1, \dots, \hat{v_j}, \dots, v_i]$$

($\hat{v_j}$: removing v_j)

$$\sum_{\sigma \in K_i} \alpha_\sigma \sigma \longmapsto \sum \alpha_\sigma \partial_i(\sigma)$$

Q Using $\mathbb{Z}$ (see why (-1) matters) or $\mathbb{Z}_2$ (used often, simplex is there or not.) coef, compute boundaries of all simpices in:



Q For $\mathbb{Z}_2$ coef, what are all of the i-chains for $i = \{0$

Q Is C_i a group?

$\ldots, \hat{v}_j, \ldots, v_i]$

↑ removing v_j

(σ)

note: σ is a chain!

$$\sigma = 1\cdot\sigma + \sum_{\tau \in K_i \mid \sigma} 0\cdot\tau$$

$\sigma \in K_i$ is an i-simplex

aka, $\sigma = \{v_0, v_1, \ldots, v_i\}$ is a set of $(i+1)$ vertices.

an oriented simplex is a set of $(i+1)$ vertices in an order where $\sigma_1 \sim \sigma_2$ iff they differ by an even permutation

$[0,1] \not\sim [1,0]$ differ by 1 transposition

$[0,1,2] \sim [2,0,1] \sim [1,2,0]$ rep σ

$\neq$

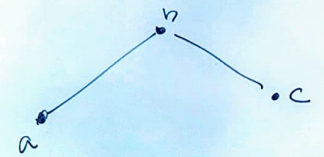
$[0,2,1] \sim [2,1,0) \sim [1,0,2]$ rep $(-\sigma)$

Q For $\mathbb{Z}_2$ coef, what are all of the i-chains for $i = \{0,1,2,3,\ldots\}$?

Q Is C_i a group?

$$\partial([v_0, v_1]) = (-1)^0 v_1 + (-1)^1 v_0$$
$$= v_1 - v_0$$

$$\partial([v_1, v_0]) = v_0 - v_1$$



$$\partial([a,b] + [b,c]) = b - a + c - b = c - a$$

$$\partial([a,b] + [c,b]) = b - a + b - c = 2b - a - c$$



$$\partial([a,b,c])$$

$$= (-1)^0[b,c] + (-1)^1[a,c] + (-1)^2[a,b]$$

$$= [b,c] - [a,c] + [a,b]$$

Boundary:

$\partial_i : C_i$

$\sigma = [v_$

$\in K_i$

$\sum_{\sigma \in K_i} \alpha_$

Q Using

compute

in :

$K =$

$\partial_i : C_i \longrightarrow C_{i-1}$

Q - What is the ker of this? (What maps to 0?)

- What is the image/range of this?